Barn Mass Balance

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WHYSOR

Everlasting IT

```
import os, json
```

Achtergrond

In Nederland is men momenteel vooral geïnteresseerd in de emissie uit stallen met dieren. We kunnen onderscheid maken tussen 2 type stallen: de natuurlijk geventileerde stallen (open stallen) en de mechanisch geventileerde stallen (gesloten stallen). Over het algemeen zitten melkkoeien (en -geiten) in open stallen en zitten de intensievere dieren (pluimvee, varkens, kalveren) in gesloten stallen.

Emissieberekening

Emissie wordt berekend door de gemeten concentratie van een bepaalde stof te vermenigvuldigen met het ventilatiedebiet. De eenheid van emissie wordt uitgedrukt in massa/tijdseenheid" (zoals kg/uur).

Concentratiemeting

De concentratie van de stof wordt gemeten met sensoren. Deze worden op bepaalde plekken bij de luchtinlaat en luchtuitlaat van de stal gehangen. Er is een consensus dat de gemeten concentraties dan betrouwbaar en representatief zijn.

Ventilatiedebiet

Mechanisch geventileerde stallen

In mechanisch geventileerde stallen wordt het ventilatiedebiet bepaald met behulp van meetwaaiers. Deze meetwaaiers worden op ventilatoren geplaatst en kunnen meten hoeveel lucht er per tijdseenheid door de ventilator wordt geblazen. Omdat mechanisch geventileerde stallen maar één (of enkele) uitstroomopeningen hebben, kan op deze manier worden bepaald wat het totale ventilatiedebiet van een stal is. Er is een consensus dat het bepalen van ventilatiedebiet met behulp van meetwaaiers betrouwbaar en representatief is.

Natuurlijk geventileerde stallen

Natuurlijk geventileerde stallen zijn voornamelijk open, waardoor het niet mogelijk is om met meetwaaiers te bepalen wat het debiet van de stallen is. Daarom wordt bij deze stallen gebruik gemaakt van de CO_2 massabalans. Dit is een theoretische benadering gebaseerd op verschillende parameters die door de tijd heen kunnen veranderen.

Emissieberekening met CO_2 massabalans

De ratiomethode is een veelgebruikte methode om de emissie van ammoniak (NH_3) in stallen te schatten. Deze methode maakt gebruik van de concentratie van CO_2 als tracergas, omdat CO_2 een relatief constante productie heeft in de stal en goed te berekenen en meten is. De basis van de ratiomethode is het idee dat de

verhouding tussen de concentraties van CO_2 binnen en buiten de stal een indicatie geeft van het ventilatiedebiet en daarmee de totale emissie van NH_3.

Een randvoorwaarde van de ratiomethode is dat de concentraties van NH_3 en het tracergas - in dit geval CO_2 - op dezelfde meetpunten en met dezelfde meetfrequentie gemeten moeten worden. Om een goede schatting van de emissie te verkrijgen is het van belang dat de concentratieratio's per meetpunt worden geschat en daarna een gemiddeld van deze waarden wordt genomen, in plaats dan eerst een gemiddelde concentratie van al die punten te bepalen en daarna de ratiomethode te gebruiken.

Support calculations

Gas density

The density of a gas can be calculated using the ideal gas law:

$$\rho = \frac{Mp}{R \cdot T}$$

where: * M is the molar mass of the gas in kg/mol

- p is the pressure in Pascals
- T is the temperature in Kelvin
- R is the universal gas constant

---

source

gas_density

```
gas_density (P:float, T:float, ppm:float, molweight:float)
```

Calculates mass density in grams per cubic metre P : pressure in Pa T : temperature in degrees Kelvin ppm : measured parts per million molweight: molecular weight in grams per mole

	Type	Details
P	float	pressure in Pascal
T	float	temperature in Kelvin
ppm	float	measured parts per million
molweight	float	molecular weight in grams per mole

Gasconcentraties in de lucht

Concentraties van chemicaliën in de lucht worden meestal gemeten als de massa van chemicaliën (milligram, microgram, nanogram of picogram) per volume lucht (kubieke meter of kubieke voet). Concentraties kunnen ook worden uitgedrukt als delen per miljoen (ppm) of delen per miljard (ppb) door gebruik te maken van een conversiefactor. Deze conversiefactor is gebaseerd op het moleculair gewicht van de chemische stof en is voor elke chemische stof verschillend. Typisch worden conversies voor chemicaliën in de lucht gemaakt met een veronderstelling van een druk van 1 atmosfeer en een temperatuur van 25 graden Celsius. Voor deze omstandigheden is de vergelijking om te converteren van concentratie in delen per miljoen naar concentratie in milligram per kubieke meter (mg/m3) als volgt:

$$\text{Concentratie (mg/m3)} = 0.0409 \times \text{concentratie (ppm)} \times \text{moleculair gewicht}$$

Concentrations of chemicals

Concentrations of chemicals in the air are usually measured as the mass of chemicals (milligrams, micrograms, nanograms or picograms) per volume of air (cubic meters or cubic feet). Concentrations can also be expressed as parts per million (ppm) or parts per billion (ppb) by using a conversion factor. This conversion factor is based on the molecular weight of the chemical and it is different for every chemical. The temperature of the atmosphere also has an influence on the calculation.

Typically, conversions for chemicals in air are made as suming a pressure of 1 atmosphere and a temperature of 25 degrees Celsius. For these conditions, the equation to convert from concentration in parts per million to con centration in milligrams per cubic meter (mg/m3) is as follows:

$$\text{Concentration (mg/m3)} = 0.0409 \times \text{concentration (ppm)} \times \text{molecular weight}$$

Functie implementatie

---

source

gas_density_from_sensor_measurment

```
gas_density_from_sensor_measurment (ppm:float, molweight:float)
```

Calculates mass density in milligrams per cubic metre

	Type	Details
ppm	float	measured parts per million
molweight	float	molecular weight in grams per mole

CO_2 productie

De CO_2 productie in een stal (in m^3 / uur) kan worden berekend met behulp van de volgende formules voor melkvee en pinken

Melkvee

$$PCO_2 = 0.2\frac{5.6m^{0.75} + 22Y_1 + 1.6 \times 10^{-5}p^3}{1000}$$

Where:

- m is the live weight in kg
- Y_1 is the daily milk production in kg per dier per dag
- p number of dracht dagen

Functie implementatie

---

source

PCO2_melkvee

```
PCO2_melkvee (aantal, melkproductie, drachtdagen, gewicht)
```

CO2 productie van melkvee per dier per dag gewicht: (gemiddelde) gewicht van de dieren melkproductie: melkproductie in kg per dier per dag drachtdagen: gemiddelde drachttijd (in dagen) De defaults zijn voor droogstaande koeien

	Details
aantal	number of animals
melkproductie	milk production in kg per animal per day
drachtdagen	days carrying (average)
gewicht	average weight of the animals in kg

```
test_args_melkvee = dict(
    aantal=130,
    melkproductie=28,
    drachtdagen=160,
    gewicht=650
)

PCO2_melkvee(**test_args_melkvee)
```

```
np.float64(36.46325050213528)
```

```
test_args_droog = dict(
    aantal=6,
    melkproductie=0,
```

```
    drachtdagen=220,
    gewicht=650
)

PCO2_melkvee(**test_args_droog)
```

```
np.float64(1.0695176539447053)
```

Pinken

$$PCO_2 = 0.2\frac{7.64m^{0.69} + Y_2\left(\frac{23}{M} - 1\right)\left(\frac{57.27+0.302m}{1-0.171Y_2}\right) + 1.6 \times 10^{-5}p^3}{1000}$$

Where:

- m is the live weight in kg
- M is the energy content of their food in MJ per kg
- Y_2 is the daily weight gain in kg per dier per dag
- p number of dracht dagen

Functie implementatie

---

source

PCO2_pinken

```
PCO2_pinken (aantal, energievoeding, drachtdagen, gewicht,
             gewichtstoename)
```

CO2 productie van pinken

	Details
aantal	number of animals
energievoeding	energy feed
drachtdagen	days carrying (average)
gewicht	average weight of the animals in kg
gewichtstoename	average weight gain of the animals in kg per day

```
test_args_pinken = dict(
    aantal=0,
    energievoeding=10,
    drachtdagen=140,
    gewicht=400,
    gewichtstoename=0.6
)
PCO2_pinken(**test_args_pinken)
```

```
np.float64(0.0)
```

```
test_args_pinken_niet_drachtig = dict(
    aantal=0,
    energievoeding=10,
    drachtdagen=0,
    gewicht=250,
    gewichtstoename=0.6
)
PCO2_pinken(**test_args_pinken_niet_drachtig)
```

```
np.float64(0.0)
```

Temperatuur correctie

Temperatuur heeft invloed op spijsveetering en gedrag en daarmee op de CO_2 productie, correctie kan worden toegepast met de volgende formule:

$$PCO_2(T) = PCO_2 \times (1000 + 4 \times (20 - T_{stal}))/1000$$

---

source

PCO2_temperatuurcorrectie

```
PCO2_temperatuurcorrectie (pco2, temperatuur)
```

Bereken temperatuur correctie voor de CO2 productie

	Details
pco2	calculated CO2 production in cubic meters per hour
temperatuur	temperature in the barn in degrees Celsius

Parameters

Voor het CO2-productiemodel zijn een aantal productiegegevens nodig. Melkproductie en –samenstelling worden altijd gemeten en gerapporteerd. De andere benodigde parameters (diergewicht, dagen in dracht, en voor jongvee de energiewaarde van het voer en gewichtstoename), worden bij voorkeur op basis van metingen op de bedrijfslocaties vastgesteld. Wanneer deze niet beschikbaar zijn dienen de volgende standaardwaarden voor te worden gebruikt.

```
pd.DataFrame(_default_parameters).set_index('categorie')
```

categorie	gewicht	drachtdagen	energievoeding	gewichtstoe-name
melkvee	650	160	NaN	NaN
droogstaande koeien	650	220	NaN	NaN
drachtig jongvee	400	140	10.0	0.6
niet drachtig jongvee	250	0	10.0	0.6

```
print(json.dumps(default_pco2_parameters, indent=4))
```

```
{
    "melkvee": {
        "gewicht": 650,
        "drachtdagen": 160
    },
    "droogstaande koeien": {
        "gewicht": 650,
        "drachtdagen": 220
    },
    "drachtig jongvee": {
        "gewicht": 400,
        "drachtdagen": 140,
        "energievoeding": 10.0,
        "gewichtstoename": 0.6
    },
    "niet drachtig jongvee": {
        "gewicht": 250,
        "drachtdagen": 0,
        "energievoeding": 10.0,
        "gewichtstoename": 0.6
    }
}
```

---

source

create_pco2_function_mapping

```
create_pco2_function_mapping (pco2_parameters)
```

Create a mapping of category to PCO2 calculation functions

---

source

PCO2_calculation

```
PCO2_calculation (mapping, category, aantal, **kwargs)
```

Test berekeningen

Emissie berekeningen

Ratiomethode

De ammoniakemissies (E_i; in kg/jaar per dierplaats) worden per meetdag i bepaald op basis van de geschatte CO_2 - productie in de stal (PCO_{2i}; in m^3 CO_2 /uur), en de gemiddelde concentratieratio van CO_2 en NH_3 als CR_i over alle meetpunten m waar CO_2 - en NH_3 concentraties tegelijkertijd in de stal gemeten zijn:

$$E_i = PCO_{2i} \cdot CR_i$$

Voor CR_i

$$CR_i = \frac{1}{m} \sum_m \frac{(NH_3)_{im}^{stal} - \left(\overline{NH_3}\right)_i^{buiten}}{(CO_2)_{im}^{stal} - \left(\overline{CO_2}\right)_i^{buiten}}$$

$$\overline{X_i^{buiten}} = \sum_m X_i^{buiten}$$

Waarin

$\boldsymbol{X_{im}^{stal}}$
: het 24-uurs gemiddelde van de concentratie van stof X in stal op meetdag i en op meetpunt m

$\boldsymbol{X_{im}^{buiten}}$
: het 24-uurs gemiddelde van de concentratie van stof X in de ingaande lucht op meetdag i en op meetpunt m

$\overline{\boldsymbol{X_i^{buiten}}}$
: het 24-uurs gemiddelde van de concentratie van stof X in de ingaande lucht op meetdag i over alle meetpunten

$$\overline{(NH_3)_i}^{buiten} = \frac{1}{m} \sum_m (NH_3)_i^{buiten}$$

and

$$\overline{CO_{2i}}^{buiten} = \frac{1}{n} \sum_{k=1}^{n} (CO_2)_{ik}^{buiten}$$

Ratiomethode met twee meetpunten

Wanneer er slechts twee meetpunten zijn, een binnen en een buiten, dan vervalt de berekening van de gemiddelden over meetpunten en kan de emissie worden berekend met de vereenvoudiging van CR_i:

$$CR_i = \frac{(NH_3)_i^{stal} - (NH_3)_i^{buiten}}{(CO_2)_i^{stal} - CO_{2i}^{buiten}}$$

en

$$E_i = PCO_{2i} \cdot CR_i$$

wordt

$$E_i = PCO_{2i} \cdot \frac{(NH_3)_i^{stal} - (NH_3)_i^{buiten}}{(CO_2)_i^{stal} - CO_{2i}^{buiten}}$$

Implementatie ratiomethode

We verwachten dat de gebruiker de volgende data als timeseries dataframe aanlevert:

Kolomnaam	Omschrijving	Eenheid
CO2_stal	CO2 concentratie in de stal in ppm	ppm
CO2_buiten	CO2 concentratie buiten de stal in ppm	ppm
NH3_stal	NH3 concentratie in de stal in ppm	ppm
NH3_buiten	NH3 concentratie buiten de stal in ppm	ppm
temperatuur	Temperatuur in de stal in Celcius	°C

Daarnaast verwachten we dat de gebruiker de volgende gegevens meegeeft

bezetting
: aantal dieren in de stal per categorie als dictionary met categorie als key en aantal als value

parameters
: dictionary met de parameters voor de verschillende categorieën. Missende waardes voor parameters worden aangevult uit de volgende standaard parameters:

```
pd.DataFrame(default_pco2_parameters).transpose()
```

	gewicht	drachtdagen	energievoeding	gewichtstoename
melkvee	650.0	160.0	NaN	NaN
droogstaande koeien	650.0	220.0	NaN	NaN
drachtig jongvee	400.0	140.0	10.0	0.6
niet drachtig jongvee	250.0	0.0	10.0	0.6

Externe data

Data voor verificatie van de implementatie wordt veelal aangeleverd in excel werkboeken. Deze data kan worden ingelezen en aangepast aan onze behoeften.

VERA data

```
vera_data_filename = os.path.join(os.getcwd(), '..', 'data', 'massabalans', 'Rekenbestand
emissie VERA.xlsx')
vera_dataframe = pd.read_excel(vera_data_filename, sheet_name='Emissions (daily means)',
header=3, index_col=7, parse_dates=True).dropna(axis=1, how='all')
```

```
vera_dataframe
```

	Measurement Institute	Animal Category	Housing system	Measurement location	Measurement period	Measurement year	Day-time period (in period)	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Cow-milk production rate > 25 >= 90%	C3: heifers < 30%	Sum-milk	C1: cows	C2: Cow-milk production rate >25.1 >= 80%	C3: pro-duction	C4: urea in milk > 15	C5: cows < 25%	Sum-dry.1
Date																					
2011-04-04	Institute 1	Cattle	Housing system 1	Location 1	1	94	13.5	76.0	17.9	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-04-05	Institute 1	Cattle	Housing system 1	Location 1	2	95	13.5	76.0	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0

Date	Measurement institute	Animal Category	Housing system	Measurement location	Measurement period	Measurement year	Day in period (in period)	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk production > 25	Summary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production > 25.1	C4: urea content > 15	C5: dry cows < 25% in milk	Summary.1
2011-04-06	Institute 1	Cattle 1	Housing system 1	Location 1	1	3	96	13.5	76.0	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-06-06	Institute 1	Cattle 1	Housing system 1	Location 1	2	1	157	13.5	76.0	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-06-07	Institute 1	Cattle 1	Housing system 1	Location 1	2	2	158	13.5	76.0	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-06-08	Institute 1	Cattle 1	Housing system 1	Location 1	2	3	159	13.4	77.0	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-08-02	Institute 1	Cattle 1	Housing system 1	Location 1	3	1	214	13.4	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-08-03	Institute 1	Cattle 1	Housing system 1	Location 1	3	2	215	13.4	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0

	Measurement institute	Animal Category	House ing system	Measurement location	Measurement period	Measurement day (in period)	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk production > 25	Sum-milk	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production > 25.1	C4: urea > 15	C5: cows < 25% in milk	Sum-dry.1
Date																					
2011-08-01	Institute 1	Cattle	House ing system 1	Location 1	3	3	216	13.4	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-10-01	Institute 1	Cattle	House ing system 1	Location 1	4	1	279	13.4	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-10-01	Institute 1	Cattle	House ing system 1	Location 1	4	2	280	13.4	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-10-01	Institute 1	Cattle	House ing system 1	Location 1	4	3	281	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-11-01	Institute 1	Cattle	House ing system 1	Location 1	5	1	328	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-11-01	Institute 1	Cattle	House ing system 1	Location 1	5	2	329	13.3	77.0	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0

Measurement	Animal Category	Institute	House ing system	Measurement period	Measurement year	Measurement day (in period)	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk production > 25	Summary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production > 25.1	C4: urea content > 15	C5: cows in milk < 25%	Summary.1
Date																					
2011-11-01	Institute 1	Cattle 1	House ing system 1 1	5	3	330	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2012-01-01	Institute 1	Cattle 1	House ing system 1 1	6	1	24	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2012-01-01	Institute 1	Cattle 1	House ing system 1 1	6	2	25	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2012-01-01	Institute 1	Cattle 1	House ing system 1 1	6	3	26	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-05-01	Institute 1	Cattle 1	House ing system 2 1	1	1	131	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-07-01	Institute 1	Cattle 1	House ing system 2 1	2	1	187	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0

Date	Measurement Institute	Animal Category	House ing system	Measurement location	Measurement period	Measurement year	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk production > 25	Summary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production > 25.1	C4: urea content > 15	C5: cows in milk < 25%	Summary.1
2011-09-07	Institute 1	Cattle	Housing system 1	3	1	2	50	13.3	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-10-26	Institute 1	Cattle	Housing system 1	4	1	2	99	13.2	77.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-12-08	Institute 1	Cattle	Housing system 1	5	1	3	42	13.2	78.0	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2012-02-16	Institute 1	Cattle	Housing system 1	6	1	4	7	13.2	NaN	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-05-03	Institute 1	Cattle	Housing system 1	1	1	2	123	NaN	NaN	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-06-27	Institute 1	Cattle	Housing system 1	2	1	3	178	NaN	NaN	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0

Date	Measurement institute	Animal Category	House-ing system	Measurement location	Measurement period	Measurement year	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk production > 25	Summary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production > 25.1	C4: urea content > 15	C5: cows in milk < 25%	Summary.1
2011-08-31	Institute 1	Cattle	Housing system 1	Location 3	1	243	NaN	NaN	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-11-02	Institute 1	Cattle	Housing system 1	Location 4	1	306	NaN	NaN	17.8	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-12-15	Institute 1	Cattle	Housing system 1	Location 5	1	349	NaN	NaN	17.7	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2012-02-14	Institute 1	Cattle	Housing system 1	Location 6	1	45	NaN	NaN	17.6	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-04-12	Institute 1	Cattle	Housing system 1	Location 1	1	102	NaN	NaN	17.6	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-06-15	Institute 1	Cattle	Housing system 1	Location 2	1	166	NaN	NaN	17.5	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0

	Measurement institute	Animal Category	House-ing system	Measurement location	Measurement period	Day (in year)	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk cow production rate > 25	Sum-mary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production rate >= 25.1	C4: urea content > 15	C5: dry cows < 25% in milk	Sum-mary.1
Date																				
2011-08-17	Institute 1	Cattle	House-ing system 1	3	1	229	NaN	NaN	17.4	...	1.0	0.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
2011-10-12	Institute 1	Cattle	House-ing system 1	4	1	285	NaN	NaN	17.3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-12-01	Institute 1	Cattle	House-ing system 1	5	1	335	NaN	NaN	17.3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2012-01-31	Institute 1	Cattle	House-ing system 1	6	1	31	NaN	NaN	17.3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-04-04	Institute 1	Cattle	House-ing system 2	1	1	94	NaN	NaN	17.3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-04-05	Institute 1	Cattle	House-ing system 2	1	2	95	NaN	NaN	17.2	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	Measurement institute	Animal Category	House-ing sys-tem	Measure-ment lo-ca-tion	Measure-ment pe-riod	Measure-ment day (in pe-riod)	Day in year	Outside temper-a-ture [oC]	Outside RH [%]	Inside temper-a-ture [oC]	...	C1: cows >= 70%	C2: Oc-cu-pa-tion rate >= 90%	C3: milk pro-duc-tion	Sum-mary	C1: heifers < 30%	C2: Oc-cu-pa-tion rate >= 80%	C3: milk pro-duc-tion	C4: urea con-tent > 25.1	C5: in milk > 15	Sum-mary.1
Date																					
2011-04-01	Institute 1	Cattle 1	Housing system 1 2	Location 1	3	96	NaN	NaN	17.3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-06-01	Institute 1	Cattle 1	Housing system 1 2	Location 2	1	157	NaN	NaN	17.3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-06-01	Institute 1	Cattle 1	Housing system 1 2	Location 2	2	158	NaN	NaN	17.4	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-06-01	Institute 1	Cattle 1	Housing system 1 2	Location 2	3	159	NaN	NaN	17.5	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-08-01	Institute 1	Cattle 1	Housing system 1 2	Location 3	1	214	NaN	NaN	17.5	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-08-01	Institute 1	Cattle 1	Housing system 1 2	Location 3	2	215	NaN	NaN	17.6	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	Measurement institute	Animal Category	House-ing system	Measurement location	Measurement period	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk production rate > 25	Summary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk production rate > 25.1	C4: urea content > 15	C5: dry cows < 25% in milk	Summary.1
Date																				
2011-08-01	Institute 1	Cattle	House-ing system 1	3	3	216	NaN	NaN	17.6	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-10-01	Institute 1	Cattle	House-ing system 1	4	1	279	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-10-01	Institute 1	Cattle	House-ing system 1	4	2	280	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-10-01	Institute 1	Cattle	House-ing system 1	4	3	281	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-11-24	Institute 1	Cattle	House-ing system 1	5	1	328	NaN	NaN	17.8	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-11-25	Institute 1	Cattle	House-ing system 1	5	2	329	NaN	NaN	17.8	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	Measurement Institute	Animal Category	House-ing sys-tem	Measure-ment lo-ca-tion	Measure-ment pe-riod	Measurement (in day pe-riod)	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk pro-duction rate > 25	Sum-mary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk pro-duction rate >= 25.1	C4: urea con-tent > 15	C5: cows in milk < 25%	Sum-mary.1
Date																					
2011-11-01	Institute 1	Cattle 1	Housing system 1 2	Location 1	5	3	330	NaN	NaN	17.8	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2012-01-01	Institute 1	Cattle 1	Housing system 1 2	Location 1	6	1	24	NaN	NaN	17.8	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2012-01-01	Institute 1	Cattle 1	Housing system 1 2	Location 1	6	2	25	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2012-01-01	Institute 1	Cattle 1	Housing system 1 2	Location 1	6	3	26	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-05-05	Institute 1	Cattle 1	Housing system 2 2	Location 1	1	1	131	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-07-06	Institute 1	Cattle 1	Housing system 2 2	Location 1	2	1	187	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	Measurement Institute	Animal Category	House-ing sys-tem	Measurement location	Measurement period	Measurement year	Day in year	Outside temperature [oC]	Outside RH [%]	Inside temperature [oC]	...	C1: cows >= 70%	C2: Occupation rate >= 90%	C3: milk cu-pro-duction rate > 25	Sum-mary	C1: heifers < 30%	C2: Occupation rate >= 80%	C3: milk cu-pro-duction rate >= 25.1	C4: urea con-tent > 15	C5: cows in milk < 25%	Sum-mary.1
Date																					
2011-09-07	Institute 1	Cattle	House-ing sys-tem 2	3	1	250	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-10-26	Institute 1	Cattle	House-ing sys-tem 2	4	1	299	NaN	NaN	17.8	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2011-12-08	Institute 1	Cattle	House-ing sys-tem 2	5	1	342	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2012-02-16	Institute 1	Cattle	House-ing sys-tem 2	6	1	47	NaN	NaN	17.7	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Exctractie van werkbare data uit gegeven werkboeken

Warmte & CO_2 data

```
data = vera_dataframe.copy()
datacolumns = set(data.columns)
```

---

source

find_production_column_names

```
find_production_column_names (data:pandas.core.frame.DataFrame)
```

Find the column names for the co2 production columns in the VERA data

```
print(json.dumps(find_production_column_names(vera_dataframe), indent=3))
```

```
{
   "drachtdagen": [
      "Days in pregnancy (dry cows)",
      "Days in pregnancy (milking cows)",
      "Days in pregnancy (heifers)"
   ],
   "energievoeding": [
      "Energy value of feed (heifers; MJ/kg dry matter)"
   ],
   "melkproductie": [
      "Milk production [kg/animal/day]"
   ],
   "gewichtstoename": [
      "Weight gain heifers [kg/day]"
   ],
   "gewicht": [
      "Weight heifers (not pregnant) [kg]",
      "Weight dry cows [kg]",
      "Weight milking cows [kg]",
      "Weight heifers (pregnant) [kg]"
   ]
}
```

```
print(json.dumps(default_pco2_parameters, indent=4))
```

```
{
    "melkvee": {
        "gewicht": 650,
        "drachtdagen": 160
    },
    "droogstaande koeien": {
        "gewicht": 650,
        "drachtdagen": 220
    },
    "drachtig jongvee": {
        "gewicht": 400,
        "drachtdagen": 140,
        "energievoeding": 10.0,
        "gewichtstoename": 0.6
    },
    "niet drachtig jongvee": {
        "gewicht": 250,
        "drachtdagen": 0,
        "energievoeding": 10.0,
        "gewichtstoename": 0.6
    }
}
```

source

extract_production_column_names

```
extract_production_column_names (data:pandas.core.frame.DataFrame)
```

Extract column names for the co2 production columns from the DataFrame

	Type	Details
data	DataFrame	DataFrame with measurement data
Returns	dict	

```
print(json.dumps(extract_production_column_names(vera_dataframe), indent=4))
#extract_production_column_names(vera_dataframe)
```

```
{
    "melkvee": {
        "gewicht": [
            "Weight milking cows [kg]"
        ],
        "drachtdagen": [
            "Days in pregnancy (milking cows)"
        ]
    },
    "droogstaande koeien": {
        "gewicht": [
            "Weight dry cows [kg]"
        ],
        "drachtdagen": [
            "Days in pregnancy (dry cows)"
        ]
    },
    "drachtig jongvee": {
        "gewicht": [
            "Weight heifers (not pregnant) [kg]",
            "Weight heifers (pregnant) [kg]"
        ],
        "drachtdagen": [
            "Days in pregnancy (heifers)"
        ],
        "energievoeding": [
            "Energy value of feed (heifers; MJ/kg dry matter)"
        ],
        "gewichtstoename": [
            "Weight gain heifers [kg/day]"
        ]
    },
    "niet drachtig jongvee": {
        "gewicht": [],
        "energievoeding": [
            "Energy value of feed (heifers; MJ/kg dry matter)"
        ],
        "gewichtstoename": [
            "Weight gain heifers [kg/day]"
```

```
        ]
    }
}
```

Emissie data

---

source

find_emission_column_names

```
find_emission_column_names (data:pandas.core.frame.DataFrame)
```

Find column names for NH3, CO2 and temperature from the DataFrame

```
print(json.dumps(find_emission_column_names(vera_dataframe), indent=3))
```

```
{
   "nh3": [
      "NH3 inside [mg/m3]",
      "NH3 outside [mg/m3]",
      "NH3 Emission [kg/year per animal place]"
   ],
   "co2": [
      "CO2 inside [ppm]",
      "CO2 outside [ppm]"
   ],
   "temp": [
      "Outside temperature [oC]",
      "Inside temperature [oC]",
      "Total heat production corrected for temperature (hpu)"
   ],
   "rh": [
      "Outside RH [%]",
      "Inside RH [%]"
   ],
   "wind": []
}
```

---

source

extract_emission_column_names

```
extract_emission_column_names (data:pandas.core.frame.DataFrame)
```

Extract column names for NH3, CO2 and temperature from the DataFrame

	Type	Details
data	DataFrame	DataFrame with measurement data

	Type	Details
Returns	dict	

```
column_mapping = extract_emission_column_names(vera_dataframe)
```

```
print(json.dumps(column_mapping, indent=2)) #extract_column_names(vera_dataframe)
```

```
{
  "binnen": {
    "nh3": [
      "NH3 inside [mg/m3]"
    ],
    "co2": [
      "CO2 inside [ppm]"
    ],
    "temp": [
      "Inside temperature [oC]"
    ],
    "rh": [
      "Inside RH [%]"
    ],
    "wind": []
  },
  "buiten": {
    "nh3": [
      "NH3 outside [mg/m3]"
    ],
    "co2": [
      "CO2 outside [ppm]"
    ],
    "temp": [
      "Outside temperature [oC]"
    ],
    "rh": [
      "Outside RH [%]"
    ],
    "wind": []
  }
}
```

```
[col for loc, measures in column_mapping.items() for measure, cols in measures.items() for
col in cols]
```

```
['NH3 inside [mg/m3]',
 'CO2 inside [ppm]',
 'Inside temperature [oC]',
 'Inside RH [%]',
 'NH3 outside [mg/m3]',
 'CO2 outside [ppm]',
 'Outside temperature [oC]',
 'Outside RH [%]',
 'Winddirection',
 'Windspeed (10 m height) [m/s]']
```

```
flatten_column_mapping(column_mapping)
```

```
['NH3 inside [mg/m3]',
 'CO2 inside [ppm]',
 'Inside temperature [oC]',
 'Inside RH [%]',
 'NH3 outside [mg/m3]',
 'CO2 outside [ppm]',
 'Outside temperature [oC]',
 'Outside RH [%]',
 'Winddirection',
 'Windspeed (10 m height) [m/s]']
```

```
vera_dataframe[flatten_column_mapping(column_mapping)]
```

Date	NH3 inside [mg/m3]	CO2 inside [ppm]	Inside temperature [oC]	Inside RH [%]	NH3 outside [mg/m3]	CO2 outside [ppm]	Outside temperature [oC]	Outside RH [%]	Winddirection	Windspeed (10 m height) [m/s]
2011-04-3	3.9470204063		17.9	72.0	0	578.0	13.5	76.0	NaN	NaN
2011-04-3	3.9570204062		17.8	71.0	0	576.0	13.5	76.0	NaN	NaN
2011-04-4	4.0639857061		17.8	71.0	0	571.0	13.5	76.0	NaN	NaN
2011-06-3	3.9670204060		17.8	70.0	0	569.0	13.5	76.0	NaN	NaN
2011-06-4	4.0739857059		17.8	71.0	0	570.0	13.5	76.0	NaN	NaN
2011-06-3	3.9870204056		NaN	NaN	0	568.0	13.4	77.0	NaN	NaN
2011-08-3	3.9270204056		17.7	72.0	0	568.0	13.4	77.0	NaN	NaN
2011-08-3	3.9370204056		17.7	72.0	0	568.0	13.4	77.0	NaN	NaN
2011-08-3	3.9470204057		17.7	72.0	0	565.0	13.4	77.0	NaN	NaN
2011-10-4	4.0639857057		17.7	72.0	0	566.0	13.4	77.0	NaN	NaN
2011-10-3	3.9770204054		17.7	72.0	0	566.0	13.4	77.0	NaN	NaN
2011-10-3	3.9870204058		17.7	72.0	0	565.0	13.3	77.0	NaN	NaN
2011-11-3	3.2470204058		17.7	73.0	0	566.0	13.3	77.0	NaN	NaN
2011-11-3	3.2570204051		17.8	73.0	0	565.0	13.3	77.0	NaN	NaN
2011-11-3	3.2670204051		17.7	72.0	0	561.0	13.3	77.0	NaN	NaN
2012-01-3	3.2470204054		17.7	72.0	0	563.0	13.3	77.0	NaN	NaN
2012-01-3	3.2570204059		17.7	71.0	0	563.0	13.3	77.0	NaN	NaN
2012-01-3	3.2670204055		17.7	72.0	0	565.0	13.3	77.0	NaN	NaN

Date	NH3 inside [mg/m3]	CO2 inside [ppm]	Inside temperature [oC]	Inside RH [%]	NH3 outside [mg/m3]	CO2 outside [ppm]	Outside temperature [oC]	Outside RH [%]	Wind direction	Wind speed (10 m height) [m/s]
2011-05-3	3.97	1020	1058	17.7	72.0	0	568.0	13.3	77.0	NaN	NaN
2011-07-3	3.96	1020	1058	17.7	72.0	0	567.0	13.3	77.0	NaN	NaN
2011-09-3	3.97	1020	1058	17.7	72.0	0	569.0	13.3	77.0	NaN	NaN
2011-10-3	3.26	1020	1061	17.7	73.0	0	568.0	13.2	77.0	NaN	NaN
2011-12-3	3.98	1020	1058	17.7	73.0	0	570.0	13.2	78.0	NaN	NaN
2012-02-3	3.96	1020	1059	17.7	73.0	0	574.0	13.2	NaN	NaN	NaN
2011-05-3	3.93	1020	1059	17.7	72.0	0	575.0	NaN	NaN	NaN	NaN
2011-06-3	3.27	1020	1061	17.8	72.0	0	573.0	NaN	NaN	NaN	NaN
2011-08-3	3.37	1020	1062	17.8	72.0	0	574.0	NaN	NaN	NaN	NaN
2011-11-3	3.92	1020	1063	17.8	71.0	0	574.0	NaN	NaN	NaN	NaN
2011-12-3	3.95	1020	1063	17.7	71.0	0	571.0	NaN	NaN	NaN	NaN
2012-02-3	3.94	1020	1068	17.6	71.0	0	570.0	NaN	NaN	NaN	NaN
2011-04-3	3.82	3089	1070	17.6	71.0	0	568.0	NaN	NaN	NaN	NaN
2011-06-3	3.75	1246	1069	17.5	71.0	0	566.0	NaN	NaN	NaN	NaN
2011-08-3	3.67	9159	1070	17.4	70.0	0	565.0	NaN	NaN	NaN	NaN
2011-10-1	NaN	NaN	1069	17.3	71.0	0	563.0	NaN	NaN	NaN	NaN
2011-12-1	NaN	NaN	1067	17.3	71.0	0	563.0	NaN	NaN	NaN	NaN
2012-01-3	NaN	NaN	1062	17.3	71.0	0	561.0	NaN	NaN	NaN	NaN
2011-04-0	NaN	NaN	1059	17.3	71.0	0	560.0	NaN	NaN	NaN	NaN
2011-04-0	NaN	NaN	1047	17.2	72.0	0	561.0	NaN	NaN	NaN	NaN
2011-04-0	NaN	NaN	1034	17.3	72.0	0	558.0	NaN	NaN	NaN	NaN
2011-06-0	NaN	NaN	1027	17.3	73.0	0	555.0	NaN	NaN	NaN	NaN
2011-06-0	NaN	NaN	1016	17.4	73.0	0	553.0	NaN	NaN	NaN	NaN
2011-06-0	NaN	NaN	1017	17.5	73.0	0	555.0	NaN	NaN	NaN	NaN
2011-08-0	NaN	NaN	1017	17.5	73.0	0	556.0	NaN	NaN	NaN	NaN
2011-08-0	NaN	NaN	1018	17.6	73.0	0	555.0	NaN	NaN	NaN	NaN

Date	NH3 inside [mg/m3]	CO2 inside [ppm]	Inside temperature [oC]	Inside RH [%]	NH3 outside [mg/m3]	CO2 outside [ppm]	Outside temperature [oC]	Outside RH [%]	Wind direction	Windspeed (10 m height) [m/s]
2011-08-04	NaN	1019	17.6	73.0	0	554.0	NaN	NaN	NaN	NaN
2011-10-06	NaN	1026	17.7	NaN	0	551.0	NaN	NaN	NaN	NaN
2011-10-07	NaN	1027	17.7	NaN	0	554.0	NaN	NaN	NaN	NaN
2011-10-08	NaN	1028	17.7	NaN	0	553.0	NaN	NaN	NaN	NaN
2011-11-24	NaN	1030	17.8	NaN	0	554.0	NaN	NaN	NaN	NaN
2011-11-25	NaN	1031	17.8	NaN	0	558.0	NaN	NaN	NaN	NaN
2011-11-26	NaN	1035	17.8	NaN	0	559.0	NaN	NaN	NaN	NaN
2012-01-24	NaN	1045	17.8	NaN	0	557.0	NaN	NaN	NaN	NaN
2012-01-25	NaN	1048	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN
2012-01-26	NaN	1060	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN
2011-05-11	NaN	1066	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN
2011-07-06	NaN	1076	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN
2011-09-07	NaN	1073	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN
2011-10-26	NaN	1079	17.8	NaN	0	NaN	NaN	NaN	NaN	NaN
2011-12-08	NaN	1077	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN
2012-02-16	NaN	1079	17.7	NaN	0	NaN	NaN	NaN	NaN	NaN

Opschonen van data

Een randvoorwaarde van de ratiomethode is dat de concentraties van NH_3 en het tracergas - in dit geval CO_2 - op dezelfde meetpunten en met dezelfde meetfrequentie gemeten moeten worden. Om een goede schatting van de emissie te verkrijgen is het van belang dat de concentratieratio's per meetpunt worden geschat en daarna een gemiddeld van deze waarden wordt genomen, in plaats dan eerst een gemiddelde concentratie van al die punten te bepalen en daarna de ratiomethode te gebruiken.

We verwachten dat de metingen van veschillende sensoren komen en op verschillende tijdstippen zijn gedaan. Om te kunnen rekenen moeten rijen volledig gevult zijn. We kunnen dit doen door de data te resamplen op een vast tijdsinterval (bijv. 10 minuten).

source

resample_data

```
resample_data (data:pandas.core.frame.DataFrame, interval:str,
               method:str)
```

Resample data to a specified interval and interpolate missing values with the givien method

	Type	Details
data	DataFrame	DataFrame with measurement data
interval	str	resampling interval (e.g. '10min' for 10 minutes)
method	str	resampling method (e.g. 'linear', 'cubic')
Returns	DataFrame	

CO_2 productie

Emissie ratio

source

calculate_emission_ratio

```
calculate_emission_ratio (NH3_stal, NH3_buiten, CO2_stal, CO2_buiten)
```

Calculate the emission ratio

	Details
NH3_stal	NH3 concentration in the barn in mg/m3
NH3_buiten	NH3 concentration outside in mg/m3
CO2_stal	CO2 concentration in the barn in ppm
CO2_buiten	CO2 concentration outside in ppm

With ratios calculated we can calculate the NH_3 emission.

Uiteindelijke berekenening module

source

calculate_emission

```
calculate_emission (data:pandas.core.frame.DataFrame,
                    pco2_parameters:dict, bezetting:dict,
                    interpolate:dict={'interval': '7min', 'method':
                    'linear'})
```

Calculate the emission using the ratio method

Type	Default	Details
data DataFrame		DataFrame with measurement data
pco2_parameters dict		parameters for the PCO2 calculation
bezetting dict		dictionary with the animal categories and their counts
interpolate dict	{'interval': '7min', 'method': 'linear'}	resampling interval and method

Airflow from CO2

```
def calculate_airflow_from_co2(
        PCO2,           # CO2 production in kg per uur
        CO2_stal,      # CO2 concentration in the barn in ppm
        CO2_buiten,    # CO2 concentration outside in ppm
    ):
    '''Calculate the airflow from CO2 concentrations and production'''

    return PCO2 * 1e-6 *  CO2_buiten / CO2_stal  # m3 per uur
```

Analyse worksheet berekeningen

NH3 Emissie berekend [kg/dpl/jaar]

```
IF(
    ISNUMBER(BE4),
    BE4*(AO4-AP4)/1000000*24*365/(Q4-Y4),
    ""
)
```

$$BG_i = BE_i \times \frac{(AO_i - AP_i)}{Q_i - Y_i} \times \frac{24 \times 365}{1000000}$$

- BE_i is Debiet berekend [m3/uur]
- AO_i is NH3 concentratie stal [mg/m3]

- AP_i is NH3 concentratie buiten [mg/m3]
- Y_i is Afgedekte ligboxen
- Q_i is Dierplaatsen

Debiet berekend [m3/uur]

```
IF(
    OR(
        BD4="",AM4="*",AM4=""
    ),
    IF(
        ISNUMBER(AQ4),
        AQ4,
        ""
    ),
    BD4*0.2/(0.000001*(AM4-AN4)))
```

$$BE_i = BD_i \times \frac{0.2}{0.000001 \times (AM_i - AN_i)}$$

- BD_i is Warmteproductie (totaal, gecorrigeerd door temperatuur)
- AM_i is CO2 stal [ppm]
- AN_i is CO2 buiten [ppm]
- AQ_i is Debiet gemeten [m3/uur]

From these two equations we can derive:

$$BG_i = BD_i \times \frac{0.2}{0.000001 \times (AM_i - AN_i)} \times \frac{(AO_i - AP_i)}{Q_i - Y_i} \times \frac{24 \times 365}{1000000} \Leftrightarrow$$

$$BG_i = BD_i \times \frac{0.2}{AM_i - AN_i} \times \frac{(AO_i - AP_i)}{Q_i - Y_i} \times 24 \times 365 \Leftrightarrow$$

$$BG_i = 0.2 \times BD_i \times \frac{AO_i - AP_i}{AM_i - AN_i} \times \frac{24 \times 365}{Q_i - Y_i}$$

$$E_i = PCO_{2i} \cdot \frac{(NH_3)_i^{stal} - (NH_3)_i^{buiten}}{(CO_2)_i^{stal} - CO_{2i}^{buiten}}$$

Warmteproductie (totaal, gecorrigeerd door temperatuur)

```
IF(
    BC4="",
    "",
    IF(
        M4="*",
        BC4,
        BC4*(1000+4*(20-M4))/1000
    )
)
```

$$BD_i = BC_i \times \frac{1000 + 4 \times (20 - M_i)}{1000}$$

- BC_i is Warmteproductie (totaal)
- M_i is Temperatuur [°C]

Warmteproductie totaal [W]

```
=IF(SUM(AY4:BB4)=0,"",SUM(AY4:BB4))
```

$$BC_i = \sum_{j=AY}^{BB} P_j$$

* BC_i is Warmteproductie (totaal)

- P_j is Warmteproductie categorie (melkvee, droogstaande koeien, drachtig jongvee, niet drachtig jongvee)

Warmteproductie

Warmteproductie categorie melkvee

```
=IF(
    OR(R4="",Z4=""),
    "",
    (5.6*(IF(AD4="",'Input voor PCO2'!$C$5,AD4))^0.75+22*Z4+1.6*0.00001*(IF(AH4="",'Input
voor PCO2'!$D$5,AH4))^3)*R4/1000
)
```

$$P_{melkvee} = \frac{5.6(AD_i)^{0.75} + 22Z_i + 1.6 \times 10^{-5}(AH_i)^3}{1000} \times R_i$$

Where:

- AD_i is Gewicht melkvee [kg]
- AH_i is Drachtdagen melkvee [dagen]
- Z_i is Melkproductie melkvee [kg/dag]
- R_i is Aantal melkvee

Calculatie vergelijking met voorbeeld data

Standaard parameters

```
test_parameters ={
    'melkvee': {
        'drachtdagen': 160,
        'gewicht': 650,
        'melkproductie': 28
    },
    'droogstaande koeien': {
        'drachtdagen': 220,
```

```
        'gewicht': 650,
        'melkproductie': 28
    },
    'drachtig jongvee': {
        'drachtdagen': 140,
        'gewicht': 400,
        'energievoeding': 10.0,
        'gewichtstoename': 0.6
    },
    'niet drachtig jongvee': {
        'drachtdagen': 0,
        'gewicht': 250,
        'energievoeding': 10.0,
        'gewichtstoename': 0.6
    }
}
```

```
bezetting = {
    'melkvee': dict(aantal=130),
    'droogstaande koeien': dict(aantal=6),
    'drachtig jongvee': dict(aantal=0),
    'niet drachtig jongvee': dict(aantal=0)
}
```

Parameters importeren

```
test_data_filename = os.path.join(os.getcwd(), '..', 'data', 'massabalans',
'Testdata2.xlsx')
print(test_data_filename)
```

```
/home/fenke/repos/openstal/nbs/../data/massabalans/Testdata2.xlsx
```

```
test_productiegegevens = pd.read_excel(test_data_filename,
sheet_name='Bedrijfsproductiegegevens', header=0, index_col=0, parse_dates=True)
```

```
test_productiegegevens
```

Parameter	Waarde	Naam parameter in WLR rapport	Naam parameter in Slimme Stal	Hoe vaak deze waarde verandert
Aantal dierplaatsen (=aantal ligboxen)	179.000000	-	NaN	zelden
Aantal melkgevende koeien	110.000000	-	NaN	elke 3-7 dagen

Parameter	Waarde	Naam parameter in WLR rapport	Naam parameter in Slimme Stal	Hoe vaak deze waarde verandert
Aantal droogstaande koeien	13.000000	-	NaN	elke 3-7 dagen
Aantal drachtige pinken	14.000000	-	NaN	elke 3-7 dagen
Aantal niet-drachtig jongvee	15.000000	-	NaN	elke 3-7 dagen
Melkproductie (kg/koe/dag)	30.000000	Y1	NaN	elke 3 dagen
Ureumgetal (mg/100g)	16.000000	-	NaN	elke 3 dagen
Mest mest besmeurd oppervlakte (m2)	760.000000	-	NaN	zelden
Aantal ligboxen gesloten	21.000000	NaN	NaN	zelden
Gewicht melkkoe (kg)	650.000000	m	NaN	nooit
Gewicht droogstaande koe (kg)	650.000000	m	NaN	nooit
Gewicht drachtige pink (kg)	400.000000	m	NaN	nooit
Gewicht niet-drachtig jongvee (kg)	250.000000	m	NaN	nooit

Parameter	Waarde	Naam parameter in WLR rapport	Naam parameter in Slimme Stal	Hoe vaak deze waarde verandert
Dagen in dracht melkkoe	160.000000	p	NaN	nooit
Dagen in dracht droogstaande koe	220.000000	p	NaN	nooit
Dagen in dracht drachtige pink	140.000000	p	NaN	nooit
Energiewaarde voer drachtige pink (MJ/kg DS)	10.000000	M	NaN	nooit
Energiewaarde voer niet-drachtig jongevee (MJ/kg DS)	10.000000	M	NaN	nooit
Gewichtstoename drachtige pink (kg/dag)	0.600000	Y2	NaN	nooit
Gewichtstoename niet-drachtig jongvee (kg/dag)	0.600000	Y2	NaN	nooit
NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
PCO2 berekening	NaN	NaN	NaN	NaN

Parameter	Waarde	Naam parameter in WLR rapport	Naam parameter in Slimme Stal	Hoe vaak deze waarde veranderd
PCO2 melkvee	159.107598	NaN	NaN	NaN
PCO2 droogstaande koe	11.586441	NaN	NaN	NaN
PCO2 drachtige pink	9.459443	NaN	NaN	NaN
PCO2 niet-drachtig jongee	6.904258	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
PCO2 totaal	187.057740	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
Totaal aantal vee	152.000000	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
Aantal ligboxen open	158.000000	NaN	NaN	NaN

Data importeren

```
test_dataframe = pd.read_excel(test_data_filename, sheet_name='Ruwe Data (CARS)', header=1,
index_col=0, parse_dates=True)
```

```
test_dataframe
```

	NH3 concentratie stal (ppm)	NH3 concentratie buitenlucht (ppm)	CO2 concentratie stal1 (ppm)	CO2 concentratie buitenlucht1 (ppm)	CO2 concentratie stal2 (ppm)	Temperatuur stal1 (°C)	Temperatuur buitenlucht1 (°C)	Temperatuur stal2 (°C)	Temperatuur buitenlucht2 (°C)	Relatieve vochtigheid stal1 (%)	P.CO2 correlatiedatabiet (m3/dier/u)	Ventilatiedebiet (m3/u)	Ventilatiedebiet (m3/u).1	NH3 emissie (kg/dp/j)	NH3 emissie (kg/dp/j).1	NH3 emissie (kg/dp/j).2	Ventilatiedebiet (m3/u).1	NH3 emissie (kg/dp/j).1	NH3 emissie (kg/dp/j).2	NH3 emissie (kg/dp/j).3
Tijd																				
2025-07-01 00:00:00	7.04	10.65	3785.32	1107.9	1020.2	13.51	13.77	2.0	..	187.7	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1660
2025-07-01 00:01:00	7.04	10.62	3765.31	1107.8	1020.2	13.51	13.77	1.0	..	187.3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1521
2025-07-01 00:02:00	7.04	10.65	3715.31	1107.8	1020.2	13.51	13.77	1.0	..	187.3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1385
2025-07-01 00:03:00	7.04	10.66	3695.27	1107.8	1020.2	13.51	13.77	0.0	..	187.3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1288
2025-07-01 00:04:00	7.04	10.59	3705.26	1107.8	1020.1	13.51	13.77	1.0	..	187.3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1192
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2025-08-02 23:55:00	2.26	11.39	5490.10	6.6	21.3	15.0	14.97	4.5	..	187.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1528
2025-08-02 23:56:00	2.26	07.39	6489.10	6.6	21.3	15.0	14.97	4.5	..	187.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1583
2025-08-02 23:57:00	2.26	03.39	4488.10	6.6	21.3	15.0	14.97	4.5	..	187.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1592
2025-08-02 23:58:00	2.26	00.39	5486.10	6.5	21.3	15.0	14.97	5.0	..	187.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1624
2025-08-02 23:59:00	2.25	99.39	7485.10	6.6	21.3	15.0	14.97	5.0	..	187.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1593

```
test_dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 60 entries, NaT to NaT
Data columns (total 66 columns):
 #   Column
Non-Null Count  Dtype
---  ------
--------------  -----
 0   Measurement institute
```

```
60 non-null     object
 1   Animal Category
60 non-null     object
 2   Housing system
60 non-null     object
 3   Measurement location
60 non-null     object
 4   Measurement period
60 non-null     int64
 5   Measurement day (in period)
60 non-null     int64
 6   Date
60 non-null     datetime64[ns]
 7   Day in year
60 non-null     int64
 8   Outside temperature [oC]
24 non-null     float64
 9   Outside RH [%]
23 non-null     float64
 10  Inside temperature [oC]
59 non-null     float64
 11  Inside RH [%]
44 non-null     float64
 12  Winddirection
0 non-null      float64
 13  Windspeed (10 m height) [m/s]
0 non-null      float64
 14  Animal places
60 non-null     int64
 15  Milking cows
60 non-null     int64
 16  Dry cows
60 non-null     int64
 17  Heifers (pregnant)
60 non-null     int64
 18  Heifers (not pregnant)
60 non-null     int64
 19  Floor type (0: slatted floor; 1: closed floor)
60 non-null     int64
 20  Walking area per animal (m2)
60 non-null     float64
 21  Grazing (hours per day)
60 non-null     int64
 22  Closed cubicles
60 non-null     int64
 23  Milk production [kg/animal/day]
56 non-null     float64
 24  Milk [% protein]
60 non-null     float64
 25  Milk [% fat]
60 non-null     float64
 26  Urea content in milk [mg/100g]
56 non-null     float64
 27  Weight milking cows [kg]
60 non-null     int64
 28  Weight dry cows [kg]
60 non-null     int64
 29  Weight heifers (pregnant) [kg]
60 non-null     int64
 30  Weight heifers (not pregnant) [kg]
60 non-null     int64
 31  Days in pregnancy (milking cows)
60 non-null     int64
```

```
 32  Days in pregnancy (dry cows)
60 non-null     int64
 33  Days in pregnancy (heifers)
60 non-null     int64
 34  Energy value of feed (heifers; MJ/kg dry matter)
60 non-null     int64
 35  Weight gain heifers [kg/day]
60 non-null     float64
 36  CO2 inside [ppm]
60 non-null     int64
 37  CO2 outside [ppm]
52 non-null     float64
 38  NH3 inside [mg/m3]
33 non-null     float64
 39  NH3 outside [mg/m3]
60 non-null     int64
 40  Number of animals
60 non-null     int64
 41  Dairy cows (milking + dry)
60 non-null     int64
 42  % closed cubicles
60 non-null     float64
 43  Occupation rate (%)
60 non-null     float64
 44  Dairy cows (%)
60 non-null     float64
 45  Heifers vs. dairy cows (%)
60 non-null     float64
 46  Dry cows vs. dairy cows (%)
60 non-null     float64
 47  Heat production milking cows (hpu)
56 non-null     float64
 48  Heat production dry cows (hpu)
56 non-null     float64
 49  Heat production heifers (pregnant) (hpu)
60 non-null     float64
 50  Heat production heifers (not pregnant) (hpu)
60 non-null     float64
 51  Total heat production (hpu)
60 non-null     float64
 52  Total heat production corrected for temperature (hpu)
59 non-null     float64
 53  Ventilation rate [m3/h]
51 non-null     float64
 54  Ventilation rate [m3/h per animal]
51 non-null     float64
 55  NH3 Emission [kg/year per animal place]
32 non-null     float64
 56   C1:                  cows >= 70%
32 non-null     float64
 57  C2:                          Occupation rate >= 90%
32 non-null     float64
 58  C3:                                                                  milk production > 25
32 non-null     float64
 59  Summary
32 non-null     float64
 60   C1:                  heifers < 30%
32 non-null     float64
 61  C2:                          Occupation rate >= 80%
32 non-null     float64
 62  C3:                                                                  milk production > 25.1
32 non-null     float64
 63  C4:                                       urea content in milk > 15
```

```
32 non-null     float64
 64  C5:                             dry cows < 25%
32 non-null     float64
 65  Summary.1
32 non-null     float64
dtypes: datetime64[ns](1), float64(38), int64(23), object(4)
memory usage: 31.4+ KB
```

```
fmap = create_pco2_function_mapping(test_parameters)
```

```
data = test_dataframe[['NH3 concentratie stal (ppm)',
 'NH3 concentratie buiten (ppm)',
 'CO2 concentratie stal (ppm)',
 'CO2 concentratie buiten1 (ppm)',
 'CO2 concentratie buiten2 (ppm)',
 'CO2 concentratie buiten3 (ppm)',
 'Temperatuur stal ©']].dropna().copy() #resample_data(test_dataframe,
**dict(interval='7min', method='linear' ))
columnmapping = extract_column_names(data)

temperatuur = data[columnmapping['stal']['temp']].mean(axis=1)
```

```
KeyError: "None of [Index(['NH3 concentratie stal (ppm)', 'NH3 concentratie buiten (ppm)',\n
'CO2 concentratie stal (ppm)', 'CO2 concentratie buiten1 (ppm)',\n       'CO2 concentratie
buiten2 (ppm)', 'CO2 concentratie buiten3 (ppm)',\n       'Temperatuur stal ©'],\n
dtype='object')] are in the [columns]"
⊠[0;31m---------------------------------------------------------------------------⊠[0m
⊠[0;31mKeyError⊠[0m                                  Traceback (most recent call last)
Cell ⊠[0;32mIn[49], line 1⊠[0m
⊠[0;32m----> 1⊠[0m data ⊠[38;5;241m=⊠[39m
⊠[43mtest_dataframe⊠[49m⊠[43m[⊠[49m⊠[43m[⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mNH3
concentratie stal (ppm)⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m,⊠[49m
⊠[1;32m      2⊠[0m ⊠[43m ⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mNH3 concentratie buiten
(ppm)⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m,⊠[49m
⊠[1;32m      3⊠[0m ⊠[43m ⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mCO2 concentratie stal
(ppm)⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m,⊠[49m
⊠[1;32m      4⊠[0m ⊠[43m ⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mCO2 concentratie buiten1
(ppm)⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m,⊠[49m
⊠[1;32m      5⊠[0m ⊠[43m ⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mCO2 concentratie buiten2
(ppm)⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m,⊠[49m
⊠[1;32m      6⊠[0m ⊠[43m ⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mCO2 concentratie buiten3
(ppm)⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m,⊠[49m
⊠[1;32m      7⊠[0m ⊠[43m ⊠[49m⊠[38;5;124;43m'⊠[39;49m⊠[38;5;124;43mTemperatuur stal
©⊠[39;49m⊠[38;5;124;43m'⊠[39;49m⊠[43m]⊠[49m⊠[43m]⊠[49m⊠[38;5;241m.⊠[39mdropna()⊠[38;5;241m.⊠
[39mcopy() ⊠[38;5;66;03m#resample_data(test_dataframe, **dict(interval='7min',
method='linear' ))⊠[39;00m
⊠[1;32m      8⊠[0m columnmapping ⊠[38;5;241m=⊠[39m extract_column_names(data)
⊠[1;32m     10⊠[0m temperatuur ⊠[38;5;241m=⊠[39m
data[columnmapping[⊠[38;5;124m'⊠[39m⊠[38;5;124mstal⊠[39m⊠[38;5;124m'⊠[39m]
[⊠[38;5;124m'⊠[39m⊠[38;5;124mtemp⊠[39m⊠[38;5;124m'⊠[39m]]⊠[38;5;241m.⊠
[39mmean(axis⊠[38;5;241m=⊠[39m⊠[38;5;241m1⊠[39m)

File ⊠[0;32m~/repos/litany/.venv/lib64/python3.11/site-packages/pandas/core/
frame.py:4108⊠[0m, in ⊠[0;36mDataFrame.__getitem__⊠[0;34m(self, key)⊠[0m
⊠[1;32m   4106⊠[0m     ⊠[38;5;28;01mif⊠[39;00m is_iterator(key):
⊠[1;32m   4107⊠[0m         key ⊠[38;5;241m=⊠[39m ⊠[38;5;28mlist⊠[39m(key)
⊠[0;32m-> 4108⊠[0m     indexer ⊠[38;5;241m=⊠[39m ⊠[38;5;28;43mself⊠[39;49m⊠[38;5;241;43m.⊠
[39;49m⊠[43mcolumns⊠[49m⊠[38;5;241;43m.⊠
[39;49m⊠[43m_get_indexer_strict⊠[49m⊠[43m(⊠[49m⊠[43mkey⊠[49m⊠[43m,⊠[49m⊠[43m
```

```
⊠[49m⊠[38;5;124;43m"⊠[39;49m⊠[38;5;124;43mcolumns⊠[39;49m⊠[38;5;124;43m"⊠[39;49m⊠[43m)⊠
[49m[⊠[38;5;241m1⊠[39m]
⊠[1;32m   4110⊠[0m ⊠[38;5;66;03m# take() does not accept boolean indexers⊠[39;00m
⊠[1;32m   4111⊠[0m ⊠[38;5;28;01mif⊠[39;00m ⊠[38;5;28mgetattr⊠[39m(indexer,
⊠[38;5;124m"⊠[39m⊠[38;5;124mdtype⊠[39m⊠[38;5;124m"⊠[39m, ⊠[38;5;28;01mNone⊠[39;00m)
⊠[38;5;241m==⊠[39m ⊠[38;5;28mbool⊠[39m:

File ⊠[0;32m~/repos/litany/.venv/lib64/python3.11/site-packages/pandas/core/indexes/
base.py:6200⊠[0m, in ⊠[0;36mIndex._get_indexer_strict⊠[0;34m(self, key, axis_name)⊠[0m
⊠[1;32m   6197⊠[0m ⊠[38;5;28;01melse⊠[39;00m:
⊠[1;32m   6198⊠[0m     keyarr, indexer, new_indexer ⊠[38;5;241m=⊠[39m
⊠[38;5;28mself⊠[39m⊠[38;5;241m.⊠[39m_reindex_non_unique(keyarr)
⊠[0;32m-> 6200⊠[0m ⊠[38;5;28;43mself⊠[39;49m⊠[38;5;241;43m.⊠
[39;49m⊠[43m_raise_if_missing⊠[49m⊠[43m(⊠[49m⊠[43mkeyarr⊠[49m⊠[43m,⊠[49m⊠[43m
⊠[49m⊠[43mindexer⊠[49m⊠[43m,⊠[49m⊠[43m ⊠[49m⊠[43maxis_name⊠[49m⊠[43m)⊠[49m
⊠[1;32m   6202⊠[0m keyarr ⊠[38;5;241m=⊠[39m ⊠[38;5;28mself⊠[39m⊠[38;5;241m.⊠
[39mtake(indexer)
⊠[1;32m   6203⊠[0m ⊠[38;5;28;01mif⊠[39;00m ⊠[38;5;28misinstance⊠[39m(key, Index):
⊠[1;32m   6204⊠[0m     ⊠[38;5;66;03m# GH 42790 - Preserve name from an Index⊠[39;00m

File ⊠[0;32m~/repos/litany/.venv/lib64/python3.11/site-packages/pandas/core/indexes/
base.py:6249⊠[0m, in ⊠[0;36mIndex._raise_if_missing⊠[0;34m(self, key, indexer, axis_name)⊠
[0m
⊠[1;32m   6247⊠[0m ⊠[38;5;28;01mif⊠[39;00m nmissing:
⊠[1;32m   6248⊠[0m     ⊠[38;5;28;01mif⊠[39;00m nmissing ⊠[38;5;241m==⊠[39m
⊠[38;5;28mlen⊠[39m(indexer):
⊠[0;32m-> 6249⊠[0m         ⊠[38;5;28;01mraise⊠[39;00m
⊠[38;5;167;01mKeyError⊠[39;00m(⊠[38;5;124mf⊠[39m⊠[38;5;124m"⊠[39m⊠[38;5;124mNone of
[⊠[39m⊠[38;5;132;01m{⊠[39;00mkey⊠[38;5;132;01m}⊠[39;00m⊠[38;5;124m] are in the
[⊠[39m⊠[38;5;132;01m{⊠[39;00maxis_name⊠[38;5;132;01m}⊠[39;00m⊠[38;5;124m]⊠
[39m⊠[38;5;124m"⊠[39m)
⊠[1;32m   6251⊠[0m     not_found ⊠[38;5;241m=⊠[39m ⊠[38;5;28mlist⊠[39m(ensure_index(key)
[missing_mask⊠[38;5;241m.⊠[39mnonzero()[⊠[38;5;241m0⊠[39m]]⊠[38;5;241m.⊠[39munique())
⊠[1;32m   6252⊠[0m     ⊠[38;5;28;01mraise⊠[39;00m
⊠[38;5;167;01mKeyError⊠[39;00m(⊠[38;5;124mf⊠[39m⊠[38;5;124m"⊠[39m⊠[38;5;132;01m{⊠[39;00mnot_found⊠[38;5;132;01m}⊠
[39;00m⊠[38;5;124m not in index⊠[39m⊠[38;5;124m"⊠[39m)

⊠[0;31mKeyError⊠[0m: "None of [Index(['NH3 concentratie stal (ppm)', 'NH3 concentratie
buiten (ppm)',\n       'CO2 concentratie stal (ppm)', 'CO2 concentratie buiten1 (ppm)',\n
'CO2 concentratie buiten2 (ppm)', 'CO2 concentratie buiten3 (ppm)',\n       'Temperatuur
stal ©'],\n      dtype='object')] are in the [columns]"
```

```
columnmapping
```

```
{'stal': {'nh3': ['NH3 concentratie stal (ppm)'],
  'co2': ['CO2 concentratie stal (ppm)'],
  'temp': ['Temperatuur stal ©']},
 'buiten': {'nh3': ['NH3 concentratie buiten (ppm)'],
  'co2': ['CO2 concentratie buiten1 (ppm)',
   'CO2 concentratie buiten2 (ppm)',
   'CO2 concentratie buiten3 (ppm)'],
  'temp': []}}
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 5728 entries, 2025-08-19 00:00:00 to 2025-08-22 23:59:00
Data columns (total 7 columns):
```

```
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   NH3 concentratie stal (ppm)     5728 non-null   float64
 1   NH3 concentratie buiten (ppm)   5728 non-null   float64
 2   CO2 concentratie stal (ppm)     5728 non-null   float64
 3   CO2 concentratie buiten1 (ppm)  5728 non-null   float64
 4   CO2 concentratie buiten2 (ppm)  5728 non-null   float64
 5   CO2 concentratie buiten3 (ppm)  5728 non-null   float64
 6   Temperatuur stal ©              5728 non-null   float64
dtypes: float64(7)
memory usage: 358.0 KB
```

```
bezetting
```

```
{'melkvee': {'aantal': 130},
 'droogstaande koeien': {'aantal': 6},
 'drachtig jongvee': {'aantal': 0},
 'niet drachtig jongvee': {'aantal': 0}}
```

```
fmap.get('melkvee')(**bezetting['melkvee'])
```

```
np.float64(36.46325050213528)
```

```
PCO2_temperatuurcorrectie(
    fmap.get('niet drachtig jongvee')(**bezetting['niet drachtig jongvee']),
    temperatuur
)
```

```
Tijd
2025-08-19 00:00:00    0.0
2025-08-19 00:01:00    0.0
2025-08-19 00:02:00    0.0
2025-08-19 00:03:00    0.0
2025-08-19 00:04:00    0.0
                      ...
2025-08-22 23:55:00    0.0
2025-08-22 23:56:00    0.0
2025-08-22 23:57:00    0.0
2025-08-22 23:58:00    0.0
2025-08-22 23:59:00    0.0
Length: 5728, dtype: float64
```

```
PCO2_temperatuurcorrectie(
    fmap.get('drachtig jongvee')(**bezetting['drachtig jongvee']),
    temperatuur
)
```

```
Tijd
2025-08-19 00:00:00    0.0
2025-08-19 00:01:00    0.0
2025-08-19 00:02:00    0.0
2025-08-19 00:03:00    0.0
2025-08-19 00:04:00    0.0
                      ...
2025-08-22 23:55:00    0.0
2025-08-22 23:56:00    0.0
```

```
2025-08-22 23:57:00    0.0
2025-08-22 23:58:00    0.0
2025-08-22 23:59:00    0.0
Length: 5728, dtype: float64
```

```
PCO2_temperatuurcorrectie(
    fmap.get('droogstaande koeien')(**bezetting['droogstaande koeien']),
    temperatuur
)
```

```
Tijd
2025-08-19 00:00:00    1.063956
2025-08-19 00:01:00    1.063528
2025-08-19 00:02:00    1.063528
2025-08-19 00:03:00    1.063956
2025-08-19 00:04:00    1.063956
                         ...
2025-08-22 23:55:00    1.084063
2025-08-22 23:56:00    1.084063
2025-08-22 23:57:00    1.084063
2025-08-22 23:58:00    1.084491
2025-08-22 23:59:00    1.084063
Length: 5728, dtype: float64
```

```
PCO2_temperatuurcorrectie(
    fmap.get('melkvee')(**bezetting['melkvee']),
    temperatuur
).rename('melkvee')
```

```
Tijd
2025-08-19 00:00:00    36.273642
2025-08-19 00:01:00    36.259056
2025-08-19 00:02:00    36.259056
2025-08-19 00:03:00    36.273642
2025-08-19 00:04:00    36.273642
                         ...
2025-08-22 23:55:00    36.959151
2025-08-22 23:56:00    36.959151
2025-08-22 23:57:00    36.959151
2025-08-22 23:58:00    36.973736
2025-08-22 23:59:00    36.959151
Name: melkvee, Length: 5728, dtype: float64
```

```
pd.concat(
        [
            PCO2_temperatuurcorrectie(
                fmap.get(category)(**params)*5,
                temperatuur
            ).rename(category)
            for category, params in bezetting.items()
        ], axis=1
    ).sum(axis=1)
```

```
Tijd
2025-08-19 00:00:00    186.687989
2025-08-19 00:01:00    186.612923
2025-08-19 00:02:00    186.612923
2025-08-19 00:03:00    186.687989
```

```
2025-08-19 00:04:00    186.687989
                           ...
2025-08-22 23:55:00    190.216069
2025-08-22 23:56:00    190.216069
2025-08-22 23:57:00    190.216069
2025-08-22 23:58:00    190.291135
2025-08-22 23:59:00    190.216069
Length: 5728, dtype: float64
```

```
emissie = calculate_emission(
    data=data,
    pco2_parameters=test_parameters,
    bezetting=bezetting,
    interpolate=dict(interval='7min', method='linear')
).resample('1h').mean()
```

```
emissie / emissie.mean()
```

```
Tijd
2025-08-19 00:00:00    0.420778
2025-08-19 01:00:00    0.428174
2025-08-19 02:00:00    0.395387
2025-08-19 03:00:00    0.496850
2025-08-19 04:00:00    0.615691
                          ...
2025-08-22 19:00:00    0.489988
2025-08-22 20:00:00    0.245229
2025-08-22 21:00:00    0.267093
2025-08-22 22:00:00    0.573332
2025-08-22 23:00:00    0.595650
Freq: h, Length: 96, dtype: float64
```

```
test_dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 5760 entries, 2025-08-19 00:00:00 to 2025-08-22 23:59:00
Data columns (total 34 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   NH3 concentratie stal (ppm)       5758 non-null   float64
 1   NH3 concentratie buiten (ppm)     5758 non-null   float64
 2   CO2 concentratie stal (ppm)       5758 non-null   float64
 3   CO2 concentratie buiten1 (ppm)    5752 non-null   float64
 4   CO2 concentratie buiten2 (ppm)    5750 non-null   float64
 5   CO2 concentratie buiten3 (ppm)    5748 non-null   float64
 6   Temperatuur stal °                5752 non-null   float64
 7   Temperatuur buiten1 °             5752 non-null   float64
 8   Temperatuur buiten2 °             5750 non-null   float64
 9   Temperatuur buiten3 °             5748 non-null   float64
 10  Luchtvochtigheid stal (%)         5752 non-null   float64
 11  Luchtvochtigheid buiten1 (%)      5752 non-null   float64
 12  Luchtvochtigheid buiten2 (%)      5750 non-null   float64
 13  Luchtvochtigheid buiten3 (%)      5748 non-null   float64
 14  Temperatuur meetbuis °            5758 non-null   float64
 15  Luchtvochtigheid meetbuis (%)     5758 non-null   float64
 16  Windrichting (graden)             96 non-null     float64
 17  Windsnelheid (km/u)               96 non-null     float64
```

```
 18  NH3 concentratie stal (mg/m3)    5760 non-null   float64
 19  NH3 concentratie buiten (mg/m3)  5760 non-null   int64
 20  CO2 concentratie stal (mg/m3)    5760 non-null   float64
 21  CO2 concentatie buiten (mg/m3)   5760 non-null   float64
 22  CO2 correctie (ppm)              5760 non-null   int64
 23  CO2 correctie (mg/m3)            5760 non-null   float64
 24  PCO2 correctie                   5760 non-null   float64
 25  Ventilatiedebiet (m3/u)          5760 non-null   float64
 26  Ventilatiedebiet (m3/dier/u)     5758 non-null   float64
 27  NH3 emissie (kg/u)               5760 non-null   float64
 28  NH3 emissie (kg/j)               5760 non-null   float64
 29  NH3 emissie (kg/dp/j)            5760 non-null   float64
 30  Ventilatiedebiet (m3/u).1        5752 non-null   float64
 31  NH3 emissie (kg/u).1             5752 non-null   float64
 32  NH3 emissie (kg/j).1             5752 non-null   float64
 33  NH3 emissie (kg/dp/j).1          5752 non-null   float64
dtypes: float64(32), int64(2)
memory usage: 1.5 MB
```

```
test_dataframe
```

	NH3 concentratie stal (ppm)	NH3 concentratie buiten (ppm)	CO2 concentratie stal (ppm)	CO2 concentatie buiten (ppm)	CO2 correctie (ppm)	Temperatuur stal ©	Temperatuur buiten ©	Temperatuur stal ©	Temperatuur buiten ©	P.CO2 correctie	Ventilatiedebiet (m3/u)	Ventilatiedebiet (m3/dier/u)	NH3 emissie (kg/u)	NH3 emissie (kg/j)	NH3 emissie (kg/dp/j)	Ventilatiedebiet (m3/u).1	NH3 emissie (kg/u).1	NH3 emissie (kg/j).1	NH3 emissie (kg/dp/j).1
Tijd																			
2025-08-01 00:00:00	0.1	0.1	989	636	9.4	9.1	4.5	5.2	0.1	3.1	9.4	1	9.2	2	0.3	..	186	7.0	8.4	9.9	5.4	8.6	3.6	7.4	4.9	3.1	5.1	3	2	7.0	9
2025-08-01 00:01:00	0.1	0.1	989	837	0.4	9.4	4.5	3.2	0.1	4.1	9.4	1	9.3	2	0.3	..	186	7.0	6.8	6.2	7.3	4.2	4.6	9.0	7.3	7.6	3.1	5.1	3	2	7.9	2
2025-08-01 00:02:00	0.1	0.1	990	037	0.4	9.7	4.5	3.2	0.1	4.1	9.4	1	9.2	2	0.3	..	186	7.0	4.5	9.2	9.7	9.4	4.4	9.8	5.8	2.4	4.1	5.1	3	2	7.4	9
2025-08-01 00:03:00	0.1	0.1	990	836	8.4	9.6	4.5	5.2	0.1	3.1	9.4	1	9.2	2	0.3	..	186	6.8	7.4	8.8	4.9	4.9	4.4	6.7	4.5	7.2	4.1	5.1	3	2	8.7	2
2025-08-01 00:04:00	0.1	0.1	991	036	7.4	9.4	4.5	6.2	0.1	3.1	9.3	1	9.2	2	0.3	..	186	6.8	7.5	8.5	4.0	4.6	3.8	6.6	4.9	7.4	0.1	5.1	3	2	7.9	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2025-08-28 23:55:00	0.2	0.2	261	139	5.4	9.0	4.0	3.3	0.1	6.6	1	5.0	1	4.9	1	6.4	..	190	.2	7.6	1.2	9.8	4.2	4.8	7.2	7.6	8.4	3	0.2	5.9	7	5.0	8
2025-08-28 23:56:00	0.2	0.2	260	739	6.4	8.9	4.0	3.4	0.1	6.6	1	5.0	1	4.9	1	6.4	..	190	.2	8.0	6.8	9.5	2.3	4.8	8.1	2.4	5.8	3	0.4	2.6	9	1.7	8

	NH3 concentratie stal (ppm)	NH3 concentratie stal (mg/m3)	CO2 concentratie buiten1 (ppm)	CO2 concentratie buiten2 (ppm)	CO2 concentratie buiten3 (ppm)	CO2 concentatie buiten (mg/m3)	Temperatuur stal ©	Temperatuur buiten1 ©	Temperatuur buiten2 ©	Temperatuur buiten3 ©	P. CO2 correctiebiet (m3/u)	Ventilatiedier (m3/u)	Ventilatiedier (m3/u).1	NH3 emissie (kg/dp/j)	NH3 emissie (kg/dp/j).1	NH3 emissie (kg/dp/j).2	Ventilatiebiet (m3/u)	NH3 emissie (kg/dp/j)	NH3 emissie (kg/dp/j).1	NH3 emissie (kg/dp/j).2
Tijd																				
2025-08-22 23:57:00	39.4	40.8	884.0	887.0	6.6	15.0	14.9	16.5	..	..	190.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	15.92
2025-08-22 23:58:00	39.5	40.8	864.0	891.0	6.5	15.0	14.9	16.5	..	..	190.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	16.24
2025-08-22 23:59:00	39.7	40.8	854.0	892.0	6.6	15.0	14.9	16.5	..	..	190.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	15.93

```python
print(json.dumps(extract_column_names(test_dataframe), indent=2))
```

```
{
  "stal": {
    "nh3": [
      "NH3 concentratie stal (ppm)",
      "NH3 concentratie stal (mg/m3)"
    ],
    "co2": [
      "CO2 concentratie stal (ppm)",
      "CO2 concentratie stal (mg/m3)"
    ],
    "temp": [
      "Temperatuur stal \u00a9"
    ]
  },
  "buiten": {
    "nh3": [
      "NH3 concentratie buiten (ppm)",
      "NH3 concentratie buiten (mg/m3)"
    ],
    "co2": [
      "CO2 concentratie buiten1 (ppm)",
      "CO2 concentratie buiten2 (ppm)",
      "CO2 concentratie buiten3 (ppm)",
      "CO2 concentatie buiten (mg/m3)"
    ],
    "temp": [
      "Temperatuur buiten1 \u00a9",
      "Temperatuur buiten2 \u00a9",
      "Temperatuur buiten3 \u00a9"
    ]
  }
}
```

```
import nbdev; nbdev.nbdev_export()
```

Bibliography